THE RIEMANN HYPOTHESIS AS MOMENT RIGIDITY OVER MEIXNER–POLLACZEK POLYNOMIALS

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Abstract. Let $dm_\infty(s) = (2\pi)^{-3/2}\left|\Gamma\left(\frac{1}{4} + \frac{is}{2}\right)\right|^2 ds$ be the probability measure on $\mathbb{R}$ arising as the archimedean spectral measure in the work of Connes, Consani and Moscovici, and let $\{\mathcal{P}_n\}_{n\geq 0}$ be its orthonormal polynomials—a rescaled Meixner–Pollaczek family, with Jacobi coefficients $a_n = \frac{1}{2}\sqrt{(2n+1)(2n+2)}$. Let ζ_{on} be the modified zeta function obtained by projecting every non-trivial zero of ζ horizontally onto the critical line, and consider the discrepancy measure

$$d\nu(s) = \left(\left|\zeta(\tfrac{1}{2} + is)\right|^2 - \left|\zeta_{\mathrm{on}}(\tfrac{1}{2} + is)\right|^2\right) dm_\infty(s).$$

We prove unconditionally that ν is a finite, non-negative, even measure with all moments finite, and that the Riemann Hypothesis is equivalent to the *moment rigidity* statement: the spectral moments $\Delta_n = \int_{\mathbb{R}} \mathcal{P}_n^2\, d\nu$ are constant in n. The proof of the non-trivial direction uses no asymptotics of orthogonal polynomials and no positivity of any kernel. Constancy of Δ_n forces, by triangularity of $\{\mathcal{P}_n^2\}$ inside the even polynomials, the proportionality of all moments of ν to those of m_∞; Carleman's condition—verified in one line from $a_n = \frac{1}{2}\sqrt{(2n+1)(2n+2)}$—makes the moment problem of m_∞ determinate, whence $\nu = c\, m_\infty$ as measures; and Hardy's 1914 theorem on the existence of zeros on the critical line forces $c = 0$, hence $\nu = 0$, hence RH. The criterion is an equivalence, not a proof of RH; we state precisely what it does and does not give.

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1. Introduction

The criterion proved in this paper can be stated in one sentence. Let $d\nu = \big(|\zeta|^2 - |\zeta_{\rm on}|^2\big)\, dm_\infty$ be the discrepancy, along the critical line, between the squared modulus of ζ and that of its on-line projection $\zeta_{\rm on}$ (every zero moved horizontally onto $\operatorname{Re} s = \frac{1}{2}$; Definition 3.1), weighted by the archimedean measure $dm_\infty(s) = (2\pi)^{-3/2}|\Gamma(\frac{1}{4} + \frac{is}{2})|^2 ds$ of Connes–Consani–Moscovici. Then (Theorem 4.4)

$$\boxed{\mathrm{RH} \iff \Delta_n := \int_{\mathbb{R}} \mathcal{P}_n(s)^2\, d\nu(s) \text{ is constant in } n \geq 0,}$$

where $\{\mathcal{P}_n\}$ are the orthonormal polynomials of m_∞, i.e. (Proposition 2.3) rescaled Meixner–Pollaczek polynomials $P_n^{(1/4)}(s/2; \pi/2)$.

1.1. **Why this measure.** The measure m_∞ is not chosen for convenience. In the spectral-realization programme of Connes, Consani and Moscovici [3, 4, 5], the operator-theoretic framework built on the semilocal trace formula attaches, to each finite set S of places of $\mathbb{Q}$ containing the archimedean place, a determinate moment problem whose measure is the squared absolute value, restricted to the critical line, of the product of the local factors over S. For $S = \{\infty\}$ the local factor is the archimedean one, $\pi^{-s/2}\Gamma(s/2)$, and the resulting measure is exactly m_∞. Connes–Consani–Moscovici prove [3, Prop. 3.3] that the orthonormal polynomials of m_∞ realize the scaling operator as the Jacobi matrix with zero diagonal and off-diagonal entries $a_n = \frac{1}{2}\sqrt{(2n+1)(2n+2)}$, and they determine its moments [3, Prop. 5.2]; the deformation of this moment problem by adjoining finite places is studied in [4], and the framework is developed further in [5]. The measure m_∞ is thus the canonical archimedean reference measure of that programme, and $|\zeta(\frac{1}{2} + is)|^2\, dm_\infty(s)$ is its natural global counterpart. The present paper takes from [3] only two inputs, both re-verifiable from classical sources: the measure m_∞ itself (Lemma 2.2 re-derives its normalization) and the exact value of its Jacobi coefficients (Proposition 2.3 re-derives them from the Meixner–Pollaczek tables [7]). Everything else is proved here from scratch.

1.2. **The shape of the proof.** The implication RH $\Rightarrow$ (Δ_n constant) is trivial: under RH the projection does nothing, $|\zeta_{\rm on}| = |\zeta|$ on the critical line, $\nu = 0$, and $\Delta_n \equiv 0$. The content is the converse, and its mechanism is the rigidity of a determinate Hamburger moment problem:

(1) the squares $\{\mathcal{P}_k^2\}_{k\leq n}$ form a triangular basis of the even polynomials of degree $\leq 2n$ (Lemma 2.7), so $\Delta_n \equiv c$ forces *every* moment of ν to equal c times the corresponding moment of m_∞;
(2) Carleman's condition holds for m_∞—in Jacobi form it is the one-line verification $\sum 1/a_n \geq \sum 1/(n+1) = \infty$—so the moment problem of m_∞ is determinate and $\nu = c\, m_\infty$ *as measures* (Theorem 4.1);
(3) both measures have continuous Lebesgue densities, and by Hardy's theorem [6] ζ has at least one zero on the critical line; at such a zero the density of ν vanishes while that of $c\, m_\infty$ is $c\, w > 0$ unless $c = 0$. Hence $c = 0$, so $\nu = 0$, and the unconditional equality case of the Hadamard-product inequality (Proposition 3.4) gives RH.

No asymptotics of $\mathcal{P}_n$, no Christoffel-function estimates, and no positivity of any indefinite kernel enter the argument. The unconditional inputs are collected in Section 3: $d\nu \geq 0$ (an explicit Hadamard-product computation: each off-line zero quadruple contributes a factor ≥ 1 to $|\zeta/\zeta_{\rm on}|^2$), finiteness of all moments of ν, and parity.

1.3. **What the criterion is and is not.** The criterion is an exact reformulation of RH inside classical moment theory: *the only non-negative even measure all of whose $\mathcal{P}_n$-spectral moments are equal is a multiple of the reference measure, and arithmetic—one application of Hardy's theorem—kills the multiple.* It is not, in its present form, a strategy of proof: evaluating ν requires knowing the zeros of ζ (Remark 5.4). Its interest, we believe, lies in the exact division of labour it exhibits—moment rigidity does all the analytic work, and the arithmetic input is confined to a single point—and in the family of explicit-formula identities it suggests (Question 5.6). All statements below are unconditional theorems about ζ; none assumes RH.

2. The archimedean measure and its orthonormal polynomials

Definition 2.1. Let $w(s) := (2\pi)^{-3/2}\left|\Gamma\left(\frac{1}{4}+\frac{is}{2}\right)\right|^2$ for $s \in \mathbb{R}$, and $dm_\infty(s) := w(s)\,ds$. Since Γ has no zeros, $w(s) > 0$ for every $s \in \mathbb{R}$.

Lemma 2.2 (total mass)**.** $\int_{\mathbb{R}}\left|\Gamma\left(\frac{1}{4}+it\right)\right|^2 dt = \pi\sqrt{2\pi}$, *and consequently* m_∞ *is a probability measure.*

Proof. Let $f(x) = e^{-e^x}e^{x/4}$. Substituting $u = e^x$,

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x)e^{-itx}\,dx = \int_0^\infty e^{-u}u^{1/4-it}\,\frac{du}{u} = \Gamma\left(\tfrac{1}{4}-it\right).$$

By Plancherel's theorem $\int_{\mathbb{R}}|\widehat{f}|^2 = 2\pi\int_{\mathbb{R}}|f|^2$, so

$$\int_{\mathbb{R}}\left|\Gamma\left(\tfrac{1}{4}+it\right)\right|^2 dt = 2\pi\int_0^\infty e^{-2u}u^{-1/2}\,du = 2\pi\,\Gamma\left(\tfrac{1}{2}\right)2^{-1/2} = \pi\sqrt{2\pi}.$$

With $t = s/2$ this gives $\int_{\mathbb{R}}|\Gamma(\frac{1}{4}+\frac{is}{2})|^2\,ds = (2\pi)^{3/2}$, which is the normalization of Definition 2.1. ☐

The normalization agrees with [3, Prop. 3.3, Prop. 5.2]. Let $\{\mathcal{P}_n\}_{n\ge 0}$ be the orthonormal polynomials of m_∞: $\deg\mathcal{P}_n = n$, positive leading coefficients, $\int\mathcal{P}_m\mathcal{P}_n\,dm_\infty = \delta_{mn}$; in particular $\mathcal{P}_0 \equiv 1$. Since w is even, the classical parity property [9, §2.3] gives $\mathcal{P}_n(-s) = (-1)^n\mathcal{P}_n(s)$, the recurrence has no diagonal terms, and there are $a_n > 0$ with

$$s\,\mathcal{P}_n(s) = a_n\,\mathcal{P}_{n+1}(s) + a_{n-1}\,\mathcal{P}_{n-1}(s), \qquad a_{-1} := 0. \tag{1}$$

Proposition 2.3 (Meixner–Pollaczek identification)**.** *The orthonormal polynomials of* m_∞ *are rescaled Meixner–Pollaczek polynomials: with* $P_n^{(\lambda)}(x;\phi)$ *as in* [7, §9.7],

$$\mathcal{P}_n(s) = c_n\,P_n^{(1/4)}\left(\tfrac{s}{2};\tfrac{\pi}{2}\right) \quad \textit{for some } c_n > 0, \qquad a_n = \tfrac{1}{2}\sqrt{(2n+1)(2n+2)} \quad (n \ge 0).$$

Proof. The Meixner–Pollaczek family $P_n^{(\lambda)}(x;\phi)$ is orthogonal on $\mathbb{R}$ with respect to the weight $e^{(2\phi-\pi)x}\,|\Gamma(\lambda+ix)|^2$ [7, §9.7]. For $\lambda = \frac{1}{4}$, $\phi = \frac{\pi}{2}$, $x = \frac{s}{2}$ this weight is $|\Gamma(\frac{1}{4}+\frac{is}{2})|^2$, i.e. the weight of m_∞ up to a positive constant, so the two orthonormal families coincide up to the change of variable and positive normalizations.

For the coefficients: the recurrence [7, (9.7.3)] at $\phi = \pi/2$ reads $(n+1)P_{n+1}(x) = 2x\,P_n(x) - (n+2\lambda-1)P_{n-1}(x)$, with $P_n = P_n^{(\lambda)}(\,\cdot\,;\pi/2)$. The leading coefficient k_n satisfies $k_{n+1} = \frac{2}{n+1}k_n$, so $k_n = 2^n/n!$ and $\tilde{p}_n := \frac{n!}{2^n}P_n$ is monic, with

$$x\,\tilde{p}_n(x) = \tilde{p}_{n+1}(x) + \frac{n(n+2\lambda-1)}{4}\,\tilde{p}_{n-1}(x).$$

Hence the orthonormal recurrence in the variable x has coefficients $a_n^{\mathrm{MP}} = \frac{1}{2}\sqrt{(n+1)(n+2\lambda)}$, which for $\lambda = \frac{1}{4}$ is $\frac{1}{2}\sqrt{(n+1)(n+\frac{1}{2})}$. The change of variable $s = 2x$ doubles them: $a_n = \sqrt{(n+1)(n+\frac{1}{2})} = \frac{1}{2}\sqrt{(2n+1)(2n+2)}$. ☐

Remark 2.4. The value $a_n = \frac{1}{2}\sqrt{(2n+1)(2n+2)}$ agrees with [3, Prop. 3.3], where it is derived through the metaplectic representation; Proposition 2.3 certifies it independently against the classical tables. Consistency check at small order: (1) gives $\mu_2(m_\infty) = a_0^2 = \frac{1}{2}$ and $\mu_4(m_\infty) = a_0^2(a_0^2+a_1^2) = \frac{1}{2}\cdot\frac{7}{2} = \frac{7}{4}$, matching the moments $c_2 = \frac{1}{2}$, $c_4 = \frac{7}{4}$ computed in [3, Prop. 5.2] from the generating identity $\sum c_n(ix)^n/n! = \sqrt{2/(e^x+e^{-x})}$.

Lemma 2.5 (moment growth and determinacy)**.** *The even moments* $\mu_{2k}(m_\infty) = \int s^{2k}\, dm_\infty$ *satisfy* $\mu_{2k}(m_\infty) \le C\,\Gamma(2k+1)\,(2/\pi)^{2k}$ *for an absolute constant* C*; hence* $\mu_{2k}(m_\infty)^{1/(2k)} = O(k)$ *and*

$$\sum_{k\ge 1} \mu_{2k}(m_\infty)^{-1/(2k)} = \infty.$$

Consequently, for every $c > 0$ *the Hamburger moment problem with moment sequence* $\big(c\,\mu_j(m_\infty)\big)_{j\ge 0}$ *is determinate:* $c\,m_\infty$ *is the unique non-negative Borel measure on* $\mathbb{R}$ *with these moments* [2, 1, 8].

Proof. By Stirling's formula in the form $|\Gamma(x+iy)| \sim \sqrt{2\pi}\,|y|^{x-1/2}e^{-\pi|y|/2}$ as $|y| \to \infty$ (cf. [10, §4.42]), there is C_0 with $w(s) \le C_0\,(1+|s|)^{-1/2}e^{-\pi|s|/2}$ for all $s \in \mathbb{R}$. Hence

$$\mu_{2k}(m_\infty) \le 2C_0 \int_0^\infty s^{2k} e^{-\pi s/2}\, ds = 2C_0\,\Gamma(2k+1)\Big(\tfrac{2}{\pi}\Big)^{2k+1},$$

and $\Gamma(2k+1)^{1/(2k)} \le 2k$ gives $\mu_{2k}^{1/(2k)} \le C'k$, so the Carleman series dominates a harmonic series. For $c > 0$ the moments are $c\,\mu_{2k}$ and $c^{-1/(2k)} \to 1$, so Carleman's condition is inherited; Carleman's theorem [2], [1, Chap. II], [8, §1] gives determinacy. ☐

Remark 2.6. Equivalently, in Jacobi-coefficient form [1, Chap. I, Addenda], [8, §3]: by Proposition 2.3,

$$\sum_{n\ge 0} \frac{1}{a_n} = \sum_{n\ge 0} \frac{2}{\sqrt{(2n+1)(2n+2)}} \;\ge\; \sum_{n\ge 0} \frac{1}{n+1} = \infty,$$

so the Jacobi matrix of m_∞ is essentially self-adjoint and the moment problem is determinate. The divergence is logarithmic: the weight $e^{-\pi|s|/2}$ sits exactly at the boundary of determinacy, like $e^{-|x|}$. This borderline determinacy is the engine of the whole paper.

Lemma 2.7 (triangularity of the squares)**.** *For every* $n \ge 0$, $\operatorname{span}\{\mathcal{P}_0^2, \mathcal{P}_1^2, \ldots, \mathcal{P}_n^2\}$ *is the space of all even polynomials of degree* $\le 2n$.

Proof. Each $\mathcal{P}_k^2$ is an even polynomial (by parity of $\mathcal{P}_k$) of degree exactly $2k$ (square of a positive leading coefficient). The $n+1$ polynomials $\mathcal{P}_0^2, \ldots, \mathcal{P}_n^2$ have the distinct degrees $0, 2, \ldots, 2n$, hence are linearly independent in the $(n+1)$-dimensional space of even polynomials of degree $\le 2n$, hence span it. ☐

3. The discrepancy measure: unconditional facts

3.1. **The on-line projection of ζ.** Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ be the completed zeta function, entire of order 1 with functional equation $\xi(s) = \xi(1-s)$; its zeros are the non-trivial zeros $\rho = \beta + i\gamma$ of ζ, $0 < \beta < 1$, $\gamma \ne 0$.[1] By the functional equation and the reality of ξ on $\mathbb{R}$, the zeros off the critical line (if any) occur in quadruples

$$Q_j = \big\{\tfrac12 + \delta_j + i\gamma_j,\ \tfrac12 + \delta_j - i\gamma_j,\ \tfrac12 - \delta_j + i\gamma_j,\ \tfrac12 - \delta_j - i\gamma_j\big\}, \qquad \delta_j \in (0, \tfrac12),\ \gamma_j > 0,$$

listed with multiplicity and indexed by $j \in J$ (J finite or countable; $J = \emptyset$ iff RH). The classical bound $\sum_\rho (\frac14 + \gamma^2)^{-1} < \infty$ [10, §2.12] implies in particular $\sum_{j\in J} \gamma_j^{-2} < \infty$.

Definition 3.1 (on-line projection)**.** For $j \in J$ put

$$R_j(s) := \frac{\big((s-\frac12)^2 + \gamma_j^2\big)^2}{\big((s-\frac12)^2 - (\delta_j + i\gamma_j)^2\big)\big((s-\frac12)^2 - (\delta_j - i\gamma_j)^2\big)},$$

[1] $\gamma \ne 0$ because ζ has no zeros on the real segment $(0,1)$: for real $\sigma \in (0,1)$ the alternating series $\eta(\sigma) = \sum_{n\ge 1}(-1)^{n-1}n^{-\sigma}$ is positive, and $\zeta(\sigma) = (1 - 2^{1-\sigma})^{-1}\eta(\sigma) \ne 0$.

a rational function whose poles are exactly the quadruple Q_j and whose zeros are exactly the projections $\frac{1}{2} \pm i\gamma_j$, each double. Define

$$\xi_{\rm on}(s) := \xi(s) \prod_{j\in J} R_j(s), \qquad \left|\zeta_{\rm on}(\tfrac{1}{2}+is)\right| := \frac{\left|\xi_{\rm on}(\frac{1}{2}+is)\right|}{\left|G(\frac{1}{2}+is)\right|}, \quad G(s) := \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2),$$

which is legitimate because $|G(\frac{1}{2}+is)| = \frac{1}{2}(\frac{1}{4}+s^2)\pi^{-1/4}|\Gamma(\frac{1}{4}+\frac{is}{2})| \neq 0$ for all $s \in \mathbb{R}$.

Lemma 3.2. *The product $\prod_j R_j$ converges absolutely and locally uniformly on compact subsets of $\mathbb{C} \setminus \bigcup_j Q_j$, and $\xi_{\rm on}$ is an entire function whose zero set is exactly the multiset of projected zeros: the critical zeros of ξ (unmoved) together with the points $\frac{1}{2} \pm i\gamma_j$, $j \in J$, each with multiplicity 2 per quadruple. If RH holds then $J = \emptyset$ and $\xi_{\rm on} = \xi$. Moreover $\xi_{\rm on}(1-s) = \xi_{\rm on}(s)$, and $s \mapsto |\xi_{\rm on}(\frac{1}{2}+is)|^2$ is even and continuous on $\mathbb{R}$.*

Proof. Writing $X = (s-\frac{1}{2})^2$, a direct expansion gives

$$R_j(s) - 1 = \frac{\delta_j^2\left(2X - 2\gamma_j^2 - \delta_j^2\right)}{\left(X - (\delta_j + i\gamma_j)^2\right)\left(X - (\delta_j - i\gamma_j)^2\right)}, \tag{2}$$

so on a fixed compact set K (avoiding the finitely many Q_j that meet K) one has $|R_j(s) - 1| \leq C_K\, \delta_j^2/\gamma_j^2 \leq C_K/(4\gamma_j^2)$ for all large j, which is summable; the product therefore converges locally uniformly and defines a meromorphic function on $\mathbb{C}$ with poles exactly at the Q_j, matching the zeros of ξ there with equal multiplicity. Hence $\xi_{\rm on}$ is entire with the stated zero set. The symmetry $R_j(1-s) = R_j(s)$ is immediate (R_j depends on $(s-\frac{1}{2})^2$), and continuity of $|\xi_{\rm on}(\frac{1}{2}+is)|^2$ in s follows from local uniformity; evenness follows from $\overline{\xi(\frac{1}{2}+is)} = \xi(\frac{1}{2}-is)$ and the evenness of $R_j(\frac{1}{2}+is)$ in s. ∎

Definition 3.3. Set $\tilde{g}(s) := \left|\zeta(\frac{1}{2}+is)\right|^2 - \left|\zeta_{\rm on}(\frac{1}{2}+is)\right|^2$ and

$$d\nu(s) := \tilde{g}(s)\, dm_\infty(s) = \tilde{g}(s)\, w(s)\, ds.$$

By Lemma 3.2, $\tilde{g}$ is continuous and even on $\mathbb{R}$.

3.2. Non-negativity via the Hadamard quadruples.

Proposition 3.4 (pointwise domination; vanishing iff RH)**.** *For all $u \in \mathbb{R}$,*

$$\left|\zeta_{\rm on}(\tfrac{1}{2}+iu)\right|^2 = \left|\zeta(\tfrac{1}{2}+iu)\right|^2 \prod_{j\in J}\Big(1 + \frac{\delta_j^2}{(u-\gamma_j)^2}\Big)^{-2}\Big(1 + \frac{\delta_j^2}{(u+\gamma_j)^2}\Big)^{-2}, \tag{3}$$

with the convention that at $u = \pm\gamma_j$ both sides vanish. Consequently $\tilde{g}(s) \geq 0$ for all $s \in \mathbb{R}$, so ν is a non-negative Borel measure; and $\tilde{g} \equiv 0$ if and only if $J = \emptyset$, i.e. if and only if RH holds.

Proof. Evaluate R_j at $s = \frac{1}{2} + iu$, where $X = (s-\frac{1}{2})^2 = -u^2$. The two denominator factors are complex conjugates, so

$$R_j(\tfrac{1}{2}+iu) = \frac{(\gamma_j^2 - u^2)^2}{\left(\gamma_j^2 - \delta_j^2 - u^2\right)^2 + 4\delta_j^2\gamma_j^2} \in [0,1],$$

a non-negative real number. The key algebraic identity is

$$\left(\gamma^2 - \delta^2 - u^2\right)^2 + 4\delta^2\gamma^2 = \left[(u-\gamma)^2 + \delta^2\right]\left[(u+\gamma)^2 + \delta^2\right], \tag{4}$$

verified by expanding both sides: each equals $u^4 + \gamma^4 + \delta^4 - 2u^2\gamma^2 + 2u^2\delta^2 + 2\gamma^2\delta^2$. Hence

$$R_j(\tfrac{1}{2}+iu) = \Big(1 + \frac{\delta_j^2}{(u-\gamma_j)^2}\Big)^{-1}\Big(1 + \frac{\delta_j^2}{(u+\gamma_j)^2}\Big)^{-1} \qquad (u \neq \pm\gamma_j),$$

and $R_j(\frac{1}{2} \pm i\gamma_j) = 0$. Since $|\xi_{\mathrm{on}}(\frac{1}{2}+iu)|^2 = |\xi(\frac{1}{2}+iu)|^2 \prod_j R_j(\frac{1}{2}+iu)^2$ and the factor $|G(\frac{1}{2}+iu)|^2$ is common to ζ and ζ_{on}, identity (3) follows. Every factor of the product in (3) lies in $(0,1]$, so $|\zeta_{\mathrm{on}}(\frac{1}{2}+iu)|^2 \le |\zeta(\frac{1}{2}+iu)|^2$, i.e. $\tilde{g} \ge 0$.

If $J = \emptyset$ the product is empty and $\tilde{g} \equiv 0$. Conversely, suppose some quadruple Q_{j_0} exists and consider $u = \gamma_{j_0} + \epsilon$ for small $\epsilon > 0$ chosen so that $\zeta(\frac{1}{2}+iu) \ne 0$ (possible: zeros are isolated). Then

$$\tilde{g}(u) = \left|\zeta(\tfrac{1}{2}+iu)\right|^2 \Big(1 - \prod_j R_j(\tfrac{1}{2}+iu)^2\Big) \;\ge\; \left|\zeta(\tfrac{1}{2}+iu)\right|^2 \Big(1 - \big(1 + \tfrac{\delta_{j_0}^2}{\epsilon^2}\big)^{-2}\Big) > 0,$$

using $R_j \le 1$ for every j. Hence $\tilde{g} \not\equiv 0$. □

Remark 3.5. The intuition behind (3): pulling a zero off the critical line *increases* $|\xi|$ at every point of the line, by the factor $\big(1 + \delta^2/(u \mp \gamma)^2\big)$ per projected zero. The on-line function is the pointwise smallest member of the family obtained by sliding the zeros horizontally; RH says ξ is already minimal. The inequality $\tilde{g} \ge 0$ is unconditional precisely because it compares ξ with its own projection.

3.3. **Finiteness of all moments.**

Proposition 3.6. *For every $k \ge 0$,*

$$\mu_{2k}(\nu) = \int_{\mathbb{R}} s^{2k}\, d\nu(s) \;\le\; \int_{\mathbb{R}} s^{2k} \left|\zeta(\tfrac{1}{2}+is)\right|^2 dm_\infty(s) \;<\; \infty.$$

In particular $\nu(\mathbb{R}) < \infty$, all moments of ν are finite, all odd moments vanish, and the Carleman condition $\sum_k \mu_{2k}(\nu)^{-1/(2k)} = \infty$ holds for ν as well.

Proof. The first inequality is the pointwise bound $0 \le \tilde{g} \le |\zeta(\frac{1}{2}+is)|^2$ from Proposition 3.4. For the second, the convexity bound $|\zeta(\frac{1}{2}+it)| \ll_\varepsilon (1+|t|)^{1/4+\varepsilon}$ [10, Chap. V] and the Stirling estimate $w(s) \le C_0(1+|s|)^{-1/2} e^{-\pi|s|/2}$ (proof of Lemma 2.5) dominate the integrand by $C_\varepsilon\, s^{2k}(1+|s|)^{2\varepsilon} e^{-\pi|s|/2}$, which is integrable; as in Lemma 2.5 this also gives $\mu_{2k}(\nu) \le C\,\Gamma(2k+2)(2/\pi)^{2k}$, whence $\mu_{2k}(\nu)^{1/(2k)} = O(k)$ and the Carleman divergence. Odd moments vanish because $\tilde{g}$ and w are even. Note that no information about the configuration of hypothetical off-line zeros is needed: the domination $\tilde{g} \le |\zeta|^2$ is global. □

4. The main theorem

Throughout this section $\Delta_n := \int_{\mathbb{R}} \mathcal{P}_n^2\, d\nu$, which is finite for every n by Proposition 3.6.

Theorem 4.1 (moment rigidity)**.** *Let σ be a non-negative, even Borel measure on $\mathbb{R}$ with all moments finite, and suppose*

$$\int_{\mathbb{R}} \mathcal{P}_n(s)^2\, d\sigma(s) = c \qquad \text{for all } n \ge 0.$$

Then $\sigma = c\, m_\infty$ as Borel measures. (For $n = 0$ the hypothesis forces $c = \sigma(\mathbb{R}) \ge 0$.)

Proof. Step 1: all moments are proportional. By orthonormality $\int \mathcal{P}_n^2\, dm_\infty = 1$, so the two measures σ and $c\, m_\infty$ assign the same integral to every $\mathcal{P}_n^2$. By linearity and Lemma 2.7 they assign the same integral to every even polynomial; both assign 0 to every odd polynomial (both measures are even, and all moments converge absolutely). Hence $\mu_j(\sigma) = c\,\mu_j(m_\infty)$ for every $j \ge 0$.

Step 2: determinacy. If $c = 0$ then $\sigma(\mathbb{R}) = \mu_0(\sigma) = 0$, so $\sigma = 0 = c\, m_\infty$. If $c > 0$, the moment sequence $(c\,\mu_j(m_\infty))_j$ satisfies Carleman's condition and is therefore determinate (Lemma 2.5): it admits exactly one non-negative representing measure. Both σ and $c\, m_\infty$ represent it; hence $\sigma = c\, m_\infty$. □

Remark 4.2. Positivity of σ is essential in Step 2: determinacy is a uniqueness statement within the class of *non-negative* measures, and the conclusion fails for signed measures. This is the precise point where the unconditional positivity $d\nu \geq 0$ of Proposition 3.4 enters the chain. Note also what the theorem does *not* say: it does not force $\sigma = 0$. The measure $\sigma = m_\infty$ itself satisfies the hypothesis with $c = 1$. Separating $c\,m_\infty$ from 0 requires one further input, and that input is arithmetic.

Lemma 4.3 (Hardy kills the constant)**.** *Suppose $\nu = c\,m_\infty$ for some $c \geq 0$, with ν as in Definition 3.3. Then $c = 0$.*

Proof. The measures ν and $c\,m_\infty$ are both absolutely continuous with respect to Lebesgue measure, with densities $\tilde{g}\,w$ and $c\,w$ respectively; equality of the measures gives $\tilde{g}\,w = c\,w$ almost everywhere, hence everywhere, since both sides are continuous (Definition 3.3; w is continuous). As $w(s) > 0$ for every s, we conclude $\tilde{g}(s) = c$ for all $s \in \mathbb{R}$.

By Hardy's theorem [6] (see also [10, Thm. 10.2]), ζ has at least one zero—in fact infinitely many—on the critical line; let $\frac{1}{2} + i\gamma_*$ be one, $\gamma_* \in \mathbb{R}$. Then $|\zeta(\frac{1}{2} + i\gamma_*)|^2 = 0$, and by (3) also $|\zeta_{\mathrm{on}}(\frac{1}{2} + i\gamma_*)|^2 = 0$ (the product factor lies in $[0, 1]$). Hence $c = \tilde{g}(\gamma_*) = 0 - 0 = 0$. $\square$

Theorem 4.4 (main theorem: RH as moment rigidity)**.** *The following are equivalent:*

(i) the Riemann Hypothesis;
(ii) $\Delta_n = \int_{\mathbb{R}} \mathcal{P}_n^2\, d\nu$ is constant in $n \geq 0$;
(iii) $\displaystyle\int_{\mathbb{R}} \kappa_n(s)\, d\nu(s) = 0$ for all $n \geq 0$, where $\kappa_n := a_n^2\big(\mathcal{P}_{n+1}^2 - \mathcal{P}_n^2\big)$ and $a_n = \frac{1}{2}\sqrt{(2n+1)(2n+2)}$.

Proof. (i)$\Rightarrow$(ii): under RH, $J = \emptyset$, $\tilde{g} \equiv 0$ (Proposition 3.4), $\nu = 0$, and $\Delta_n \equiv 0$.

(ii)$\Leftrightarrow$(iii): $\int \kappa_n\, d\nu = a_n^2(\Delta_{n+1} - \Delta_n)$, all quantities being finite (Proposition 3.6) and $a_n > 0$; the equivalence is a telescoping triviality.

(ii)$\Rightarrow$(i): ν is non-negative (Proposition 3.4), even, with all moments finite (Proposition 3.6); by Theorem 4.1, $\nu = c\,m_\infty$ with $c = \Delta_0 = \nu(\mathbb{R})$; by Lemma 4.3, $c = 0$, so $\nu = 0$, i.e. $\tilde{g}\,w \equiv 0$, i.e. $\tilde{g} \equiv 0$ ($w > 0$ and $\tilde{g}$ continuous); by the equality case of Proposition 3.4, $J = \emptyset$, which is RH. $\square$

5. Remarks

Remark 5.1 (what the proof uses, and what it does not). The proof of Theorem 4.4 uses: orthonormality and parity of the $\mathcal{P}_n$; the exact Jacobi coefficients (only through Carleman's condition, where any $a_n = O(n)$ would do); the Hadamard quadruple inequality with its equality case; the convexity bound for ζ; Stirling; and Hardy's theorem. It uses no asymptotics of the polynomials $\mathcal{P}_n$, no Christoffel–Darboux kernel estimates, and no positivity of the (indefinite) functions κ_n. The division of labour is sharp: moment theory proves "ν is a multiple of m_∞"; the arithmetic of ζ enters exactly once, to prove "the multiple is zero".

Remark 5.2 (the universal quantifier is essential). No finite subfamily of the conditions in Theorem 4.4(iii) characterizes RH, and there is no single-index version. Indeed $\int \kappa_n\, dm_\infty = a_n^2(1-1) = 0$ for every n: each κ_n has mean zero against the reference measure, so each condition $\int \kappa_n\, d\nu = 0$ is a single linear constraint that non-zero non-negative measures satisfy in abundance (any truncation of the family cuts out a set of measures of finite codimension, which contains, e.g., a positive multiple of m_∞ plus suitable even perturbations). The criterion does not detect the mass of ν through the sign of any test function—it cannot, since the κ_n are indefinite—but through the *completeness* of $\{\mathcal{P}_n^2\}$ in the even polynomials combined with determinacy. Relatedly, "Δ_n constant" alone does not force $\nu = 0$ (the measure m_∞ itself has $\Delta_n \equiv 1$): it is the arithmetic nature of ν—a density vanishing at critical zeros—that converts rigidity into RH.

Remark 5.3 (partial-sum versions). It is tempting to repackage the family (iii) as a single function of a height parameter, e.g. $T_\lambda := \sum_{n:\,\gamma_n \leq \lambda} \int \kappa_n\, d\nu$, where γ_n runs over the ordinates of the zeros of ξ in non-decreasing order, and to assert RH $\iff T_\lambda = 0$ for all λ. One direction is again trivial, but the converse *as stated* recovers the per-index family only if the ordinates γ_n are pairwise distinct, since the jumps of $\lambda \mapsto T_\lambda$ then isolate each term. Under ¬RH distinctness fails automatically—$\beta + i\gamma$ and $(1-\beta)+i\gamma$ share the ordinate γ—so the λ-form only yields block sums $\sum_{n\in B} a_n^2(\Delta_{n+1}-\Delta_n) = 0$ over blocks B of coinciding ordinates, and these do not force constancy of Δ_n (one free parameter per block). We therefore regard the per-index form (ii)/(iii) as the correct formulation and state this caveat explicitly: we do not claim the λ-indexed equivalence.

Remark 5.4 (the barrier, stated honestly). The criterion is a reformulation of RH, not progress towards its proof: to evaluate ν—hence any Δ_n—one must know the multiset of zeros of ζ, since ζ_{on} is built from it. A proof of RH through Theorem 4.4 would require establishing the constancy of Δ_n by some mechanism that does not presuppose knowledge of the zeros, and no such mechanism is offered here.

Remark 5.5 (relation to the Connes–Consani–Moscovici framework). What is taken from [3, 4, 5] is the measure m_∞ and the exact value of its Jacobi coefficients, i.e. the identity of the reference moment problem; both are re-derived here from classical sources (Lemma 2.2, Proposition 2.3). The criterion of Theorem 4.4 and its proof are, to the best of the author's knowledge, not contained in those works, and conversely nothing here bears on the operator-theoretic statements proved there (semilocal trace formula, prolate operators, zeta spectral triples): the present paper makes no claim about them, in either direction. The reader should view Theorem 4.4 as a statement of classical moment theory and classical analytic number theory whose *coordinates*—the measure and the polynomials—are those canonically singled out by the Connes–Consani–Moscovici framework at the archimedean place.

Question 5.6 (arithmetic content)**.** Each function $h_n(s) := \kappa_n(s)\, w(s)$ is even, real-analytic, and decays like $e^{-\pi|s|/2}$ times a polynomial, so it is an admissible test function for the Weil explicit formula; condition (iii) of Theorem 4.4 is then a countable family of identities between a sum over the zeros and a sum over prime powers. Is there a direct arithmetic interpretation of the spectral moments Δ_n—for instance, a closed expression for the prime side $\sum_p (\log p/\sqrt{p})\, \widehat{h}_n(\log p)$ in terms of the Meixner–Pollaczek data—that would let the constancy of Δ_n be studied as a statement about primes rather than about zeros? We leave this as the natural open question raised by the criterion.

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